

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2101

II

As specified in the official PDF

Source-grounded

Fundamentals of Printing Science

Articulate the fundamental knowledge of color, human visual perception, color theories, and illumination concepts for accurate color

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2101 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2101.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2101
Course title	Fundamentals of Printing Science
Semester	II
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Electro magnetic spectrum, Newton’s color wheel, Engineering Physics for Mechanical, Structural and 2002A I/II Light, Viscosity, Industrial Applications Course Outcome Blooms Taxonomy CO Course Outcome Level Explain the principles of color and human visual perception and apply color harmony, color mixing, and CO1 Understand illumination concepts for accurate and consistent color reproduction in printing. Articulate digital materials for printing by applying basic knowledge of image quality, color reproduction, CO2 Understand and image input processes used in print production. Explain how images are reproduced with tonal variation and how print quality is evaluated and controlled CO3 Understand using basic measurement concepts. Explain how the characteristics of printing substrates and inks influence printability, machine performance, CO4 Understand and final print quality CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

33 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- Articulate the fundamental knowledge of color, human visual perception, color theories, and illumination concepts for accurate color
- understanding and reproduction in printing.
- effectively during print production.
- operation and acceptable print quality.

Course outcomes

CO	Official outcome	Study level
CO1	3 2 - 1 1 1 - - - 1	See official syllabus
CO2	3 2 1 1 2 1 - - - 1	See official syllabus
CO3	3 3 1 2 1 1 - - - 1	See official syllabus
CO4	3 2 1 1 1 2 1 1 - - 1 Total 12 9 3 5 5 5 1 1 - - 4 Correlation Level 3 2.25 1 1.25 1.25 1.25 1 1 - - 1 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Diploma Curriculum Revision 2026 2101 Page #1 Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Color: Definition. Attributes of color - Hue, Saturation, Lightness, Alliance of EM Spectrum, Wavelength of Light and Color, Human Color perception - Elements of eye, Rod	See official syllabus

CO	Official outcome	Study level
	and LMS Cone cell vision. Standard illuminant - Definition, Types of illuminants, Color temperature. Spectral	

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 - 1 1 1 - - - 1
CO2	CO2 3 2 1 1 2 1 - - - 1
CO3	CO3 3 3 1 2 1 1 - - - 1
CO4	CO4 3 2 1 1 1 2 1 1 - - 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Color Science power distribution curve 15 0	3	As specified
Module 2	Digital Imaging 15 0	7	As specified
Module 3	Halftone reproduction 15 0	11	As specified
Module 4	Direction, Porosity, Smoothness. 15 0	12	As specified

MODULE 1

Color Science power distribution curve 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Color Science power distribution curve 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ perception and Color reproduction. Maxwell Color Triangle.

perception and Color reproduction. Maxwell Color Triangle. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify perception and Color reproduction. Maxwell Color Triangle. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, perception and color reproduction. maxwell color triangle. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What perception and Color reproduction. Maxwell Color Triangle. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പ്രകാശം perception and Color reproduction. Maxwell Color Triangle. പരസ്പരം definition/meaning പരസ്പരം. പരസ്പരം പരസ്പരം പരസ്പരം, steps, application പരസ്പരം പരസ്പരം പരസ്പരം. Technical terms English-പ്രകാശം പരസ്പരം പരസ്പരം.

▼ Scanners - Working Principle & Types.

Scanners - Working Principle & Types. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Scanners - Working Principle & Types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, scanners - working principle & types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Scanners - Working Principle & Types. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- സ്കാണറുകൾ - Working Principle & Types. സ്കാണറുകളുടെ definition/meaning എന്താണ്. സ്കാണറുകൾ എങ്ങനെ പ്രവർത്തിക്കുന്നു, steps, application എങ്ങനെ സ്കാണറുകൾ ഉപയോഗിക്കുന്നു. Technical terms English-ൽ എങ്ങനെ എഴുതുന്നു.

▼ Digital image - Pixel and bit depth. Vector file formats and raster file formats.

Digital image - Pixel and bit depth. Vector file formats and raster file formats. is an official topic in Module 1 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Digital image - Pixel and bit depth. Vector file formats and raster file formats. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, digital image - pixel and bit depth. vector file formats and raster file formats. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Digital image - Pixel and bit depth. Vector file formats and raster file formats. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഡിജിറ്റൽ ഇമേജ് - പിക്സൽ ആൻഡ് ബിറ്റ് ഡിപ്പത്ത്. വെക്ടർ ഫയൽ ഫോർമാറ്റുകൾ ആൻഡ് രാസ്റ്റർ ഫയൽ ഫോർമാറ്റുകൾ. ഡിജിറ്റൽ ഇമേജിന്റെ ഡിഫിനിഷൻ/മീനിംഗ്. ഡിജിറ്റൽ ഇമേജിന്റെ ഡിഫിനിഷൻ, ഫോർമാറ്റുകൾ, സ്റ്റേപ്പുകൾ, ആപ്ലിക്കേഷൻ. ടെക്നിക്കൽ ടേർമുകൾ ഇംഗ്ലീഷിൽ എഴുതുക. Technical terms English-ൽ എഴുതുക.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Digital Imaging 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Digital Imaging 15 0
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ Digital Imaging 15 0

Digital Imaging 15 0 is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Digital Imaging 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, digital imaging 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Digital Imaging 15 0 means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഡിജിറ്റൽ ഇമേജിംഗ് 15 0 ഡിജിറ്റൽ ഇമേജിംഗ് ഡിഫിനിഷൻ/മീനിംഗ് ഡിജിറ്റൽ ഇമേജിംഗ്. ഡിജിറ്റൽ ഇമേജിംഗ് ഡിഫിനിഷൻ, steps, application ഡിജിറ്റൽ ഇമേജിംഗ് ഡിഫിനിഷൻ. Technical terms English-ൽ ഡിജിറ്റൽ ഇമേജിംഗ്.

▼ Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF,

Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, different file formats- their features, and uses in printing - jpeg/jpg, tiff, png, gif, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF, രണ്ടാം ക്ലാസ്സ് definition/meaning രണ്ടാം ക്ലാസ്സ്. രണ്ടാം ക്ലാസ്സ് രണ്ടാം ക്ലാസ്സ്, steps, application രണ്ടാം ക്ലാസ്സ് രണ്ടാം ക്ലാസ്സ് രണ്ടാം ക്ലാസ്സ്. Technical terms English- രണ്ടാം ക്ലാസ്സ് രണ്ടാം ക്ലാസ്സ്.

▼ BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD,

BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, bmp, pdf, jdf, eps, psd, crd, doc/docx, ppt/pptx, flash-swf, zip, html, qxd, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD, definition/meaning, application, steps, Technical terms English- .

▼ PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.

PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, pmd, indd, sla/scd, txt, odt, icc/icm, xml. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രിന്റിംഗ് PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML. പ്രിന്റിംഗ് പ്രക്രിയയുടെ definition/meaning വിശദീകരിക്കുക. പ്രിന്റിംഗ് പ്രക്രിയയുടെ components, steps, application വിശദീകരിക്കുക. Technical terms English-ൽ വിശദീകരിക്കുക.

▼ Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of

Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Type of Originals - Lineoriginal,Continuous tone originals, Halftone original,Areas of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, type of originals - lineoriginal,continuous tone originals, halftone original,areas of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Type of Originals - Lineoriginal, Continuous tone originals, Halftone original, Areas of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Type of Originals - Lineoriginal, Continuous tone originals, Halftone original, Areas of interest, definition/meaning, steps, application, Technical terms, English- Malayalam.

▼ Halftone screening - AM, FM & Hybrid Screening process.

Halftone screening - AM, FM & Hybrid Screening process. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Halftone screening - AM, FM & Hybrid Screening process. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, halftone screening - am, fm & hybrid screening process. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Halftone screening - AM, FM & Hybrid Screening process. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-½ Halftone screening - AM, FM & Hybrid Screening process. ഫ്ലാഷ് definition/meaning ഫ്ലാഷ്. ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്, steps, application ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്. Technical terms English-½ ഫ്ലാഷ് ഫ്ലാഷ് ഫ്ലാഷ്.

▼ Halftone dots - features and types.

Halftone dots - features and types. is an official topic in Module 2 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Halftone dots - features and types. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, halftone dots - features and types. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Halftone dots - features and types. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Halftone dots - features and types. □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Halftone reproduction 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Halftone reproduction 15 0
- Official duration: As specified
- Official topic entries captured: 11
- The full source excerpt is preserved below for coverage checking.

▼ Halftone reproduction 15 0

Halftone reproduction 15 0 is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Halftone reproduction 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, halftone reproduction 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Halftone reproduction 15 0 means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-½ Halftone reproduction 15 0

പ്രശ്നത്തിന് നിർവ്വചനം/അർത്ഥം നൽകുക. നിർവ്വചനം നൽകുന്ന
 ഘട്ടങ്ങൾ, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition.

Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, optical density - definition, plays a role in printing. tonal gradation - definition. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓപ്റ്റിക്കൽ ഡെൻസിറ്റി - ഡിഫിനിഷൻ, പ്രിന്റിംഗിൽ പങ്കു വഹിക്കുന്നു. ടോണൽ ഗ്രാഡേഷൻ - ഡിഫിനിഷൻ. ഡിഫിനിഷൻ/അർത്ഥം എഴുതുക. ഡിഫിനിഷൻ ഡിഫിനിഷൻ, steps, application എഴുതുക. ഡിഫിനിഷൻ ഡിഫിനിഷൻ. Technical terms English-ൽ എഴുതുക. ഡിഫിനിഷൻ.

▼ Halftone image - Significance and basics of halftone imaging.

Halftone image - Significance and basics of halftone imaging. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Halftone image - Significance and basics of halftone imaging. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, halftone image - significance and basics of halftone imaging. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Halftone image - Significance and basics of halftone imaging. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-½ Halftone image - Significance and basics of halftone imaging. ഫാൾട്ടോൺ ചിത്രം definition/meaning ഫാൾട്ടോൺ ചിത്രം. ഫാൾട്ടോൺ ചിത്രം ഫാൾട്ടോൺ, steps, application ഫാൾട്ടോൺ ചിത്രം ഫാൾട്ടോൺ. Technical terms English-½ ഫാൾട്ടോൺ ചിത്രം.

▼ Light sensitive material - Film (positive and negative), basics of image formation on

Light sensitive material - Film (positive and negative), basics of image formation on is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Light sensitive material - Film (positive and negative), basics of image formation on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, light sensitive material - film (positive and negative), basics of image formation on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Light sensitive material - Film (positive and negative), basics of image formation on means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-☐☐ Light sensitive material - Film (positive and negative), basics of image formation on ☐☐☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and

Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, paper and board properties: define - paper and board-sizes, classifications and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പേപ്പറും ബോർഡും Paper and Board Properties: Define - Paper and Board- Sizes, Classifications and പേപ്പറും ബോർഡും definition/meaning പേപ്പറും ബോർഡും. പേപ്പറും ബോർഡും പേപ്പറും, steps, application പേപ്പറും ബോർഡും പേപ്പറും. Technical terms English-പേപ്പറും പേപ്പറും.

▼ their uses

their uses is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify their uses using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, their uses should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What their uses means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇരുപതു their uses ഉപയോഗിക്കുന്നതിനുള്ള നിർവ്വചനം/അർത്ഥം ഉപയോഗിക്കുന്നതിനുള്ള നിർവ്വചനം. ഉപയോഗിക്കുന്നതിനുള്ള നിർവ്വചനം, steps, application ഉപയോഗിക്കുന്നതിനുള്ള നിർവ്വചനം. Technical terms English-ഇരുപതു ഉപയോഗിക്കുന്നതിനുള്ള നിർവ്വചനം.

▼ Printability & Runnability Properties.

Printability & Runnability Properties. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Printability & Runnability Properties. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, printability & runnability properties. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Printability & Runnability Properties. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രിന്റബിലിറ്റി & റണ്ണബിലിറ്റി പ്രോപ്പർട്ടീസ്. പ്രിന്റബിലിറ്റി/മീനിംഗ് പ്രിന്റബിലിറ്റി. പ്രിന്റബിലിറ്റി, steps, application പ്രിന്റബിലിറ്റി പ്രിന്റബിലിറ്റി. Technical terms English-പ്രിന്റബിലിറ്റി പ്രിന്റബിലിറ്റി.

▼ Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness.

Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, optical properties- color, gloss, opacity, metamerism, brightness, whiteness. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Optical properties- Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓപ്റ്റിക്കൽ പ്രൊപ്പർട്ടീസ്- കളർ, ഗ്ലോസ്, ഓപ്പാസിറ്റി, മെറ്റാമെറിസം, ബ്രൈറ്റ്നസ്സ്, വൈറ്റ്നസ്സ്. ഓപ്റ്റിക്കൽ പ്രൊപ്പർട്ടീസ് ന്റെ definition/meaning എന്താണ്. ഓപ്റ്റിക്കൽ പ്രൊപ്പർട്ടീസ് ന്റെ steps, application എന്താണ്. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

▼ Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire

Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, structural properties- density, basis weight and grammage, caliper and bulk, wire should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-**Structural Properties- Density, Basis Weight and Grammage, Caliper and Bulk, Wire** **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?** definition/meaning **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?**. **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?**, steps, application **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?** **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?**. Technical terms English-**എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?** **എന്തിനാണ് ഈ വിഷയം പഠിക്കേണ്ടത്?**.

▼ and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain

and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, and felt sides, moisture content and relative humidity, dimensional stability, grain should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പ്രശ്നം and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain എന്നിവയുടെ definition/meaning എഴുതുക. എന്നിവയുടെ ഘടന, steps, application എന്നിവയെക്കുറിച്ചും എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ Paper and Ink Science

Paper and Ink Science is an official topic in Module 3 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Paper and Ink Science using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, paper and ink science should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Paper and Ink Science means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പേപ്പർ and Ink Science പേപ്പർ, ഇങ്ക്. പേപ്പർ definition/meaning, ഇങ്ക് definition/meaning. പേപ്പർ, ഇങ്ക് steps, application, technical terms English- Malayalam. Technical terms English- Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Direction, Porosity, Smoothness. 15 0

This module is presented from the official syllabus scope for Fundamentals of Printing Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Direction, Porosity, Smoothness. 15 0
- Official duration: As specified
- Official topic entries captured: 12
- The full source excerpt is preserved below for coverage checking.

▼ Direction, Porosity, Smoothness. 15 0

Direction, Porosity, Smoothness. 15 0 is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Direction, Porosity, Smoothness. 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, direction, porosity, smoothness. 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Direction, Porosity, Smoothness. 15 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ in Printing Technology

in Printing Technology is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify in Printing Technology using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, in printing technology should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What in Printing Technology means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പ്രിന്റിംഗ് ടെക്നോളജി പ്രിന്റിംഗ് ടെക്നോളജി പ്രിന്റിംഗ് definition/meaning പ്രിന്റിംഗ് പ്രിന്റിംഗ്. പ്രിന്റിംഗ് പ്രിന്റിംഗ് പ്രിന്റിംഗ്, steps, application പ്രിന്റിംഗ് പ്രിന്റിംഗ് പ്രിന്റിംഗ്. Technical terms English-പ്രിന്റിംഗ് പ്രിന്റിംഗ് പ്രിന്റിംഗ്.

▼ Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding

Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mechanical properties- tear resistance, tensile strength, bursting strength, folding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-**മെക്കാനിക്കൽ ഗുണങ്ങൾ** - Tear Resistance, Tensile Strength, Bursting Strength, Folding **പ്രതിരോധശേഷി** **വിവരണം** **പ്രതിരോധശേഷി** **പ്രതിരോധശേഷി**. **പ്രതിരോധശേഷി** **പ്രതിരോധശേഷി** **പ്രതിരോധശേഷി**, steps, application **പ്രതിരോധശേഷി** **പ്രതിരോധശേഷി** **പ്രതിരോധശേഷി**. Technical terms English-**മെക്കാനിക്കൽ ഗുണങ്ങൾ** **പ്രതിരോധശേഷി**.

▼ Endurance, Stiffness, Abrasion Resistance.

Endurance, Stiffness, Abrasion Resistance. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Endurance, Stiffness, Abrasion Resistance. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, endurance, stiffness, abrasion resistance. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Endurance, Stiffness, Abrasion Resistance. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എൻഡറൻസ്, സ്റ്റിഫ്നസ്സ്, അബ്രേഷൻ റെസിസ്റ്റൻസ്. ഏതുതരത്തിലുള്ളതും ഏതുതരത്തിലുള്ളതും definition/meaning എഴുതുക. ഏതുതരത്തിലുള്ളതും ഏതുതരത്തിലുള്ളതും, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

▼ Ink properties: Optical properties - Color, Gloss, Opacity, Transparency.

Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ink properties: optical properties - color, gloss, opacity, transparency. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇൻക് പ്രൊപ്പർട്ടീസ്: ഒപ്റ്റിക്കൽ പ്രൊപ്പർട്ടീസ് - കളർ, ഗ്ലോസ്, ഓപ്പാസിറ്റി, ട്രാൻസ്പറൻസി. ഇൻക് പ്രൊപ്പർട്ടീസ് definition/meaning ഇൻക് പ്രൊപ്പർട്ടീസ്. ഇൻക് പ്രൊപ്പർട്ടീസ് പ്രൊപ്പർട്ടീസ്, steps, application ഇൻക് പ്രൊപ്പർട്ടീസ് പ്രൊപ്പർട്ടീസ്. Technical terms English-ഇൻക് പ്രൊപ്പർട്ടീസ് പ്രൊപ്പർട്ടീസ്.

▼ Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance.

Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, physical properties - length, viscosity, tack, flow, adhesion, scuff resistance. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പ്രാഥമിക ഭൗതിക ഗുണങ്ങൾ - നീളം, വിസ്കോസിറ്റി, ടാക്ക്, ഫ്ലോ, അനുചരത, സ്കാഫ് റെസിസ്റ്റൻസ്. ഈ പദങ്ങൾക്ക് ഉപയോഗിച്ച് definition/meaning നൽകുക. ഈ ഗുണങ്ങൾ എങ്ങനെ പ്രയോജനപ്പെടുന്നു, steps, application നൽകുക. Technical terms English-ൽ നൽകുക.

▼ Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances,

Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റൂറ Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, ഏകദേശം ഏകദേശം definition/meaning ഏകദേശം ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English-റൂ ഏകദേശം ഏകദേശം.

▼ viscoelasticity.

viscoelasticity. is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify viscoelasticity. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, viscoelasticity. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What viscoelasticity. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- $\square\square$ viscoelasticity. $\square\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square\square$.

▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്ന Detailed Syllabus പരീക്ഷാസൂചികയിൽ നിന്ന് definition/meaning പരീക്ഷാസൂചികയിൽ നിന്ന്. പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക, steps, application പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക. Technical terms English-ൽ പരീക്ഷാ സൂചികയിൽ പരീക്ഷാസൂചിക.

▼ Mode of

Mode of is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിവിടെ Mode of ഏകീകരണം എന്നതിന് definition/meaning എഴുതണം. ഏകീകരണം എങ്ങനെ നടത്തണം, steps, application എന്നിവയെക്കുറിച്ച് എഴുതണം. Technical terms English-ൽ എഴുതണം.

▼ Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിവിടെ Mapped Instructional ഉള്ളതിനെക്കുറിച്ച് definition/meaning ഉള്ളതുകൊണ്ട്. ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്, steps, application ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്. Technical terms English-ൽ ഉള്ളതിനെക്കുറിച്ച് ഉള്ളതിനെക്കുറിച്ച്.

▼ Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Printing Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Subtopic Student Learning Outcome delivery Taxonomy Level definition/meaning. Steps, application. Technical terms English-

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Create small design samples demonstrating: Warm color scheme Cool color scheme Monochromatic scheme Analogous scheme Complementary scheme
- Prepare a brief note explaining color and its attributes (hue, saturation, lightness) with real-life examples. 6
- Compare color appearance under daylight, tungsten, and LED light sources and prepare a report. 6 Module II
- Prepare a table listing different file formats used in printing and group formats as image, document, layout, color
- management, and workflow files. 12
- Mention their features, advantages, limitations, and typical printing applications.
- Explain the role of pixel count and bit depth in print quality. 12
- Draw a simple block diagram showing the working of a scanner. 12 Module III
- Take a photograph or printed picture and mark highlight, midtone, and shadow areas using a pencil or digital tool. Write a
- short note explaining how tonal variation appears in different areas.
- Identify and explain which screening method is most suitable for different print applications such as newspapers, packaging,
- and high-quality art prints. Module IV
- Prepare a one-page report on “Role of Paper and Ink Properties in Print Quality” with examples. 6
- Match paper types with suitable printing applications (newspaper, books, packaging, labels). 6
- Total Self-Learning hours 90 Learning Resources TEXT BOOK
- Title Author Publication
- Chemistry in Printing, 2nd Edition, 2011 Tulika Das Barnana Prakashani
- Understanding Digital Colour, 1995 P. Green GATF REFERENCE

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

– Theory						
CA3	CA4	CA1 CA5	CA6	CA2	ESE	
Mode Self-learning Written	Attendance Self-learning	Self-learning Series	Self-learning Series			
activities Exam	activities	activities Exam	activities Exam			
(Module 3) (theory)	(Module 4)	(Module 1) (2 modules)	(Module 2) (2 modules)			
Duration 2 hours	2 hours	2.5 hours				
Exam mark 15	15	15	50	50	15	60
Converted to 20	20					
Marks 20	5 60					15
Total 40 60						
Tentative End of	End of					
By 4th week			By 7th week			

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain perception and Color reproduction. Maxwell Color Triangle.. (3 marks)**

► **Define or explain Scanners - Working Principle & Types.. (3 marks)**

► **Define or explain Digital image - Pixel and bit depth. Vector file formats and raster file formats.. (3 marks)**

Module 2

► **Define or explain Digital Imaging 15 0. (3 marks)**

► **Define or explain Different File formats- their features, and uses in printing - JPEG/JPG, TIFF, PNG, GIF,. (3 marks)**

► **Define or explain BMP, PDF, JDF, EPS, PSD, CRD, DOC/DOCX, PPT/PPTX, FLASH-SWF, ZIP, HTML, QXD,. (3 marks)**

► **Define or explain PMD, INDD, SLA/SCD, TXT, ODT, ICC/ICM, XML.. (3 marks)**

Module 3

► **Define or explain Halftone reproduction 15 0. (3 marks)**

► **Define or explain Optical Density - Definition, plays a role in printing. Tonal Gradation - Definition.. (3 marks)**

► **Define or explain Halftone image - Significance and basics of halftone imaging.. (3 marks)**

► **Define or explain Light sensitive material - Film (positive and negative), basics of image formation on. (3 marks)**

Module 4

► **Define or explain Direction, Porosity, Smoothness. 15 0. (3 marks)**

► **Define or explain in Printing Technology. (3 marks)**

► **Define or explain Mechanical properties- Tear Resistance, Tensile Strength, Bursting Strength, Folding. (3 marks)**

► **Define or explain Endurance, Stiffness, Abrasion Resistance.. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **perception and Color reproduction. Maxwell Color Triangle..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Digital Imaging 15 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Halftone reproduction 15 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Direction, Porosity, Smoothness. 15 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- <https://www.colormatters.com> I
- <https://www.britannica.com> II
- <https://www.prepressure.com> III
- <https://www.paperonweb.com> IV

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 2101. Verify institutional updates against the current official curriculum.